Benchmarking Machine Learning for Chaotic Three-Body Dynamics in Astrophysics

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Abstract

Machine learning (ML) and generative AI have recently made impressive progress in understanding physics. We explore a hierarchy of prediction problems stemming from chaotic three-body systems and benchmark existing ML approaches, including general-purpose prediction algorithms and those that build on top of foundational physics models. We first simulate the trajectories of triple systems near the critical boundary of stable/unstable initial conditions. We then design a hierarchy of learning tasks, focusing mainly on two broad problems: binary stability prediction and continuous trajectory prediction. The former classifies whether a given triple system has diverged or not up to a specific horizon, while the latter predicts the exact trajectory, both given suitable input features. We benchmark a range of ML approaches, from simple regression methods to our fine-tuned state-of-the-art foundational physics models, across these three tasks, and compare them against astrophysical methods such as the Mardling–Aarseth stability criterion. Our experiments reveal a clear order of task difficulties in chaotic three-body dynamics: early dynamics enable strong stability prediction, while precise long-horizon trajectory forecasting remains difficult.

1 Introduction

Recent research in machine learning has increasingly shifted from learning the semantics of data modalities such as language and images [11, 26] toward understanding the physics underlying everyday observations, as evidenced by the growing popularity of world models [13, 14, 2] and recent physics-aware benchmarks [23, 4, 22]. In this work, we focus on one of the fundamental problems in physics that remains relatively underexplored in current AI research: the three-body problem [35, 6, 17, 36]. It is a classical problem for understanding the stability and dynamics of stellar systems, and provides one of the simplest yet best-controlled settings for benchmarking ML methods.

A triple system consists of three masses whose motions are deterministically governed by their initial physical conditions through Newtonian dynamics. Their long-term stability and orbital evolution are relevant to a broad range of astrophysical phenomena [32]. Depending on the initial conditions, a triple system may either remain stable over long timescales or rapidly diverge and become unbounded. However, between these two regimes lies a critical region in which systems that only differ very slightly in initial conditions can lead to markedly different stability outcomes [33, 36]. We focus on this boundary regime from which we select the initial conditions for our simulations, used as the ground-truth trajectories.

Given the trajectories, we design a suite of tasks with different levels of difficulty, broadly categorized as stability prediction and trajectory prediction. Stability prediction aims to determine whether or not a given triple system will remain bound or eventually become unbounded. The input features are either

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Feature	# Ch.	Physical meaning
`aratios`	1	semi-major-axis ratio $a_{\rm in}/a_{\rm out}$
`mratios`	2	mass ratios of the hierarchical triple
`eccs`	2	eccentricities e_* of the orbits
`incs`	2	inclinations of the orbital planes i_*
`APs`	4	arguments of periapsis ω_*
`LANs`	4	longitudes of ascending node Ω_*

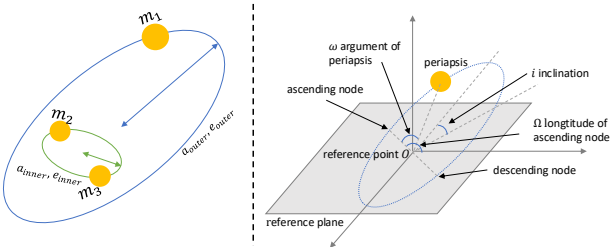


Figure 1: Orbital representation used in our experiments. $*$ stands for either the inner or the outer orbit. The rightmost panel describes features of one of the two orbits. `APs` and `LANs` are represented using their sine and cosine to avoid discontinuities at the 2π boundary.

its physical initial conditions or a short prefix of the ground-truth trajectory [17]. Trajectory prediction goes beyond the binary stability outcome and predicts the detailed dynamical evolution of the system. To see how hard these tasks are, we benchmark a range of ML approaches against conventional astrophysical methods, spanning from classical supervised learning models [29, 18, 3, 8, 9] to more recent foundation models [16, 1, 25, 15] and our own fine-tuned variants. We additionally consider established analytical stability criteria: Mardling–Aarseth [21] and Vynatheya et al. criterion [36]. Our extensive experiments show that the performance of ML methods varies depending on the task specification; while short-term and coarse-grained properties (e.g. stability) can be learned with high accuracy, long-term trajectory prediction remains substantially more challenging. Our main contributions are:

- We constructed a large-scale controlled perturbation-ensemble dataset of chaotic hierarchical three-body systems concentrated near the challenging astrophysical stability boundary.
- We design a hierarchical suite of stability and trajectory prediction tasks that probe different levels of dynamical prediction difficulty, and evaluate a broad range of ML approaches against conventional astrophysical methods.
- Through systematic experiments and analysis, we characterize both the capabilities and limitations of ML in learning chaotic three-body dynamics, providing insights into the limitations and challenges of existing ML approaches.

2 Related Work

Three-body systems with boundary conditions. Three-body systems have long been studied in the context of orbital stability. Traditional approaches rely primarily on analytical or semi-empirical criteria, such as the Eggleton–Kiseleva and Mardling–Aarseth criteria [12, 21], which characterize stability using orbital elements and mass ratios. Although computationally efficient and widely used, these criteria can become less reliable near stability boundaries or outside the regimes in which they were calibrated [33, 36]. ML methods have been applied to long-term stability prediction in both compact planetary systems and hierarchical triples. Tamayo et al. [31] combines short N-body integrations with physically motivated summary statistics to predict stability over up to 10^9 orbits. Neural networks have been used to predict instability timescales [10] and hierarchical triple-star stability [36, 37]. On trajectory prediction, Breen et al. [6] used neural networks to approximate chaotic three-body trajectories over bounded time intervals, targeting efficient trajectory prediction rather than long-term stability characterization. We evaluates both stability and trajectory predictions within a unified hierarchy of tasks, which allows us to systematically study how predictive performance changes with the available dynamical information and prediction horizon in strongly chaotic regimes.

Physics-focused benchmarks for machine learning. For broader context, physics-focused benchmarks have increasingly been used to evaluate whether ML models can simulate or predict physical dynamics. For instance, PDEBench [30], LagrangeBench [34], and The Well [24] provide standardized simulation datasets for evaluating surrogate models across diverse physical systems, with emphasis on rollout accuracy, generalization, and physically meaningful errors. Complementary benchmarks for world models and intuitive physics, including Physion [5], Physics-IQ [23], and WorldModelBench [19], examine whether learned models capture physical principles rather than merely producing visually plausible predictions. Discrete chaotic systems, such as elementary cellular

automata [7, 38], have also been used as benchmarks for studying the capabilities of neural architectures. Our benchmark complements them by providing a simple but well-controlled setting for evaluating how the predictive performance is influenced by the targets and input features of carefully constructed tasks all related by the three-body dynamics.

3 Astrophysical Setup and Learning Tasks

3.1 Astrophysical Setup and Dataset Construction

Hierarchical triple systems and controlled perturbations. Hierarchical triple systems provide a canonical setting for studying the long-term dynamics and stability of the three-body problem. We construct a controlled perturbation-ensemble dataset comprising $5{,}000$ *base triple ensembles of systems*. The initial physical conditions are 'close' to the critical stability boundary, predicted by the MLP model from Vynatheya et al. [36, 37], in phase space. Among these roughly 40% of the systems are predicted stable by this model. For each base triple system, we generate 32 variations from the same base initial conditions and a prescribed perturbation of selected initial-condition variables, yielding $160{,}000$ *individual triple systems* in total. We then employ the Newtonian *N*-body code `REBOUND` [27], using the 15th-order integrator `IAS15` [28], to compute the gravitational dynamics and the resulting trajectories. Each trajectory is sampled at $2{,}001$ time points on a common integration grid, with consecutive samples separated by five outer orbital periods.

The 32 variations associated with one base system differ only in orbital phases or slight semi major axis perturbations, and thus should not be treated as independent samples. We therefore treat each 32-realization perturbation ensemble as the fundamental unit for dataset splitting and ensemble-level evaluation. This design preserves the response of each system to controlled changes in its initial conditions and provides an empirical distribution of stability outcomes, enabling both individual-level and ensemble-level tasks, such as stable-fraction estimation and survival curve analysis. The full perturbation schedule, parameter ranges, integration settings, and preprocessing procedure are described in Appendix A.1 and A.2.

Scale-invariant processing. We represent the dynamical state using dimensionless orbital descriptors rather than raw Cartesian positions and velocities. This choice reduces sensitivity to nuisance degrees of freedom and provides a representation more directly aligned with the underlying orbital dynamics. These descriptors also provide a physically meaningful representation for the prediction problems considered in this work. Unless otherwise specified, all experiments use this orbital representation. The resulting `orbital15` feature set, summarized in Figure 1, provides a common state representation across stability-prediction and trajectory-forecasting tasks, enabling consistent comparisons across model families.

3.2 Stability of Triple-Star Systems

The overall goal of the stability prediction tasks is to determine whether a given triple-star system will remain bounded at a future timestamp, typically far from its current state. Here, we focus on several sub-tasks that predict stability based on different types of input information, as well as a survival prediction task under controlled small perturbations. In particular, we use each base system, which consists of 32 realizations as detailed in Sec. 3.1, as the minimal unit for prediction tasks.

Task A: Stability prediction from initial physical conditions. Given the initial condition $\mathbf{c}_i$ of a nominal system i, we seek to predict its long-term stability at a future horizon H. Because each nominal system is simulated with V nearby phase-space perturbations, we define its stability as the fraction of variations that remain bound until H:

$$\widehat{q}_i(H) = f_{\mathrm{A}}(\mathbf{c}_i), \qquad q_i(H) = \frac{1}{V}\sum_{j=1}^{V} \mathbb{I}(T_{ij} \geq H)\,, \tag{1}$$

f_{A} denotes the prediction model, T_{ij} is the instability time of the j-th variation of ensemble i, and $\mathbb{I}(\cdot)$ is the indicator function. Thus, $q_i(H) \in [0, 1]$ is the empirical stable fraction and $\widehat{q}_i(H)$ is its prediction. For the binary all-stable setting, the label is defined as $y_i = \mathbb{I}(q_i(H) = 1)$, where $y_i = 1$ indicates that all variations of an ensemble remain stable and $y_i = 0$ indicates that at least one variation becomes unstable. This initial-condition-based formulation is closely related to conventional analytical stability criteria [21] and enables rapid screening without performing a long-term numerical integration.

Task B: Stability prediction from a short-term trajectory prefix. Given a short-term trajectory prefix $\mathbf{X}_{ij}^{(P)} = [\mathbf{x}_{ij}(t_1), \ldots, \mathbf{x}_{ij}(t_P)] \in \mathbb{R}^{P\times d}$ for variations j of ensemble of system i, we seek to predict whether the individual system remains stable until a future horizon H, where $t_P < H$. The task is formulated as

$$\widehat{p}_{ij}(H) = f_{\mathrm{B}}\left(\mathbf{X}_{ij}^{(P)}\right), \qquad y_{ij}(H) = \mathbb{I}(T_{ij} \geq H)\,. \tag{2}$$

Here, $\mathbf{x}_{ij}(t_p) \in \mathbb{R}^d$ is the d-dimensional orbital state observed at timestamp t_p, P is the prefix length, and f_{B} outputs the predicted probability $\widehat{p}_{ij}(H)$ of survival. The binary label $y_{ij}(H)$ equals one for a stable individual and zero for an individual system that becomes unstable before H. Unlike Task A, this formulation exploits early dynamical evolution, which may reveal precursors of instability that are not apparent from the initial conditions alone [31].

Task C: Survival prediction within a ensemble of systems. Given the collection of short-term prefixes $\mathcal{X}_i = \{\mathbf{X}_{ij}^{(P)}\}_{j=1}^{V}$ from all V variations of ensemble i, we seek to predict the survival probability of each variation at multiple future horizons $H_1 < \cdots < H_K$. We parameterize the prediction using discrete conditional hazards:

$$\begin{aligned} \widehat{h}_{ij,k} &= \Pr(H_{k-1} < T_{ij} \leq H_k \mid T_{ij} > H_{k-1},\, \mathcal{X}_i)\,, \\ \widehat{S}_{ij}(H_k) = \prod_{\ell=1}^{k}\left(1-\widehat{h}_{ij,\ell}\right), &\qquad \widehat{Q}_i(H_k) = \frac{1}{V}\sum_{j=1}^{V}\widehat{S}_{ij}(H_k). \end{aligned} \tag{3}$$

$\widehat{h}_{ij,k}$ is the predicted conditional probability that variation j becomes unstable during the interval $(H_{k-1}, H_k]$, while $\widehat{S}_{ij}(H_k)$ is its predicted probability of remaining stable beyond H_k. $\widehat{Q}_i(H_k)$ denotes the predicted surviving fraction within ensemble i, K is the number of prediction horizons, and $V = 32$ in our experiments. The corresponding empirical target is $Q_i(H_k) = \frac{1}{V}\sum_{j=1}^{V}\mathbb{I}(T_{ij} > H_k)$. This formulation produces a monotonic survival curve rather than a single binary outcome, while naturally representing trajectories that remain stable throughout the finite simulation horizon.

3.3 Trajectory Forecasting of Detailed Dynamics

In addition to the stability of the entire ensemble of system, we also investigate whether the existing ML methods can predict the fine-grained orbital evolution of chaotic three-body systems, characterized by a series of trajectory physical quantities. Specifically, we evaluate whether state-of-the-art pretrained time-series models can predict the future orbital evolution of chaotic three-body systems from a finite observation history. Given a 512-step context represented by `orbital15` features as mentioned in Figure 1, the model predicts the subsequent dynamics. We consider both a one-shot 128-step forecast and a longer 512-step open-loop rollout obtained by recursively applying the same predictor over four consecutive 128-step blocks without access to future observations. As orbital evolution is conventionally studied through numerical integration [35, 28], this task provides a natural test of whether pretrained time-series models can serve as data-driven predictors of chaotic dynamics.

4 Experiments and Analysis

4.1 Experimental Setup

Baseline methods. The methods used in our experiments can be mainly classified into three categories, spanning domain-specific astrophysical criteria, task-specific supervised learning models, and recent pretrained time-series foundation models. **1. Physics-based empirical criteria:** We include classical stability criteria from celestial mechanics, such as the Mardling–Aarseth criterion [21] and Vynatheya et al. criterion [36], which estimate the stability of hierarchical three-body systems using analytically motivated empirical relations between orbital and physical parameters. **2. Task-specific supervised learning models:** We evaluate a range of standard machine learning and deep learning approaches trained specifically for each prediction task. These include feature-based models such as XGBoost, as well as neural architectures including multilayer perceptrons (MLP), convolutional neural networks (CNN), temporal convolutional networks (TCN), and attention-based models. Depending on the task, these models operate either on engineered dynamical summaries(A.2) or directly on orbital trajectories. **3. Pretrained time-series foundation models:** We further evaluate recent large-scale time-series models pretrained on diverse temporal data, as well as our fine-tuned variants. In particular, we consider models capable of multivariate sequence modeling and assess their ability to transfer general temporal representations to chaotic three-body dynamics, both directly and, where applicable, through task-specific adaptation.

Table 1: Representative results across the benchmark. Full results are reported in Appendix A.5.

Prediction setting	Representative method	Key result	Main finding
Initial-condition stability	XGBoost	Binary: AUROC 0.8525, Fraction: MAE 0.2330	Partially predictable
Prefix-based stability	Dynamical summaries + XGBoost	Binary: AUROC 0.9851, Fraction: MAE 0.0398	Early dynamics highly informative
Multi-horizon survival	Chronos-2 + direct pooling	MAE 0.0376, IBS 0.0416	Strong survival prediction
Trajectory forecasting	Chronos-2 (zero-shot)	MAE 0.228	Competitive short-term forecasting

Evaluation metrics. We use task-specific evaluation metrics throughout the benchmark. For classification tasks, we primarily report accuracy, F1 score, AUROC, and AUPRC. Regression tasks are evaluated using MAE, MSE, RMSE, R^2, and correlation coefficients. For survival prediction, we report horizon-wise MAE and integrated Brier score(IBS). For trajectory forecasting, we use MSE and MAE in the standardized feature space, together with correlation and unit-circle error for angular variables. We additionally report horizon-wise errors for long-horizon rollouts to quantify error accumulation over time. Details of dataset splits can be found in Appendix A.3. More implementation details are included in Appendix A.4.

4.2 Main Results and Analysis

Stability prediction from initial physical conditions. Initial physical conditions provide substantial but incomplete, information about long-term stability. For binary classification, learned models consistently outperform the analytical stability criteria. XGBoost achieves the highest accuracy 0.805 and F1 score 0.535, compared with 0.619 accuracy and 0.441 F1 for the Mardling–Aarseth criterion [21]. Meanwhile, the MLP classifier from Vynatheya et al. [36] achieves the strongest threshold-independent discrimination, with an AUROC of 0.909 and an AUPRC of 0.718. For stable-fraction regression, XGBoost obtains the lowest MSE 0.088 and highest R^2 0.444, while the the MLP classifier from Vynatheya et al. [36] achieves a slightly lower MAE 0.218 vs. XGBoost 0.233. Overall, these results show that initial orbital parameters contain meaningful predictive information beyond fixed analytical criteria, but still leave substantial uncertainty about long-term outcomes.

Early dynamics are highly informative for stability prediction. Short trajectory prefixes contain strong signals of future stability. Using dynamical summary statistics, XGBoost achieves AUROCs of 0.965 and 0.985 with 96- and 512-step prefixes, respectively, consistently outperforming models trained directly on raw sequences. Frozen PANDA [16] and Chronos-2 [1] representations are also competitive, with Chronos-2 reaching an AUROC of 0.979 and an AUPRC of 0.955 at 512 steps. For the more demanding multi-horizon survival task, pretrained encoders achieve the strongest performance: the best Chronos-2 model obtains an overall MAE of 0.0376 and an IBS of 0.0416, compared with 0.310 and 0.132 for the train-mean baseline. These results suggest that early dynamical evolution contains strong precursors of long-term instability, while the most effective representation depends on the prediction objective: physically motivated dynamical summaries excel at binary classification, whereas pretrained sequence representations are particularly effective for multi-horizon survival prediction.

Long-horizon trajectory forecasting remains challenging. Zero-shot foundation models already capture substantial dynamical structure in three-body trajectories, achieving performance comparable to the supervised multivariate iTransformer [20] despite no task-specific training. For example, PANDA [16] obtains an MSE of 0.222, close to the supervised iTransformer at 0.212, while Chronos-2 [1] achieves the lowest MAE 0.228 and unit-circle error 0.227 . This suggests that pretrained temporal representations can transfer effectively to chaotic orbital dynamics. Nevertheless, prediction errors accumulate consistently during the 512-step open-loop rollout across all models. Such error accumulation represents a key limitation of current learning-based forecasters and becomes particularly severe in chaotic systems, where small prediction errors are progressively amplified over time. Thus, while foundation models exhibit encouraging zero-shot dynamical prediction ability, reliable fine-grained forecasting over long horizons remains an open challenge.

5 Conclusion and Discussions

We introduced a benchmark for machine learning on chaotic three-body dynamics near the astrophysical stability boundary. Our results show that stability becomes more predictable with early dynamical observations, while physically motivated features and pretrained representations remain competitive. For trajectory forecasting, foundation models can match supervised models, but long-horizon prediction remains challenging.

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A Technical appendices and supplementary material

A.1 Construction of Initial Conditions and Variations

The initial conditions have 32 phase-space variations of 5,000 base triple systems. Each triple consists of 3 point masses present in two nested binary orbits. The 'inner' binary consists of the first two point masses, and the 'outer' binary consists to the third point mass and the total inner binary mass. The initial physical parameter distributions are as follows:

- Three masses m_1 and m_2, and m_3 are uniformly sampled, spanning 2 orders of magnitude. This results in two mass ratios m_1/m_2 (≤ 1) and $m_3/(m_1 + m_2)$.
- Two semimajor axes $a_{\rm in}$ and $a_{\rm out}$ are uniformly sampled, spanning 4 orders of magnitude. This results in one semimajor axis ratio $a_{\rm in}/a_{\rm out}$ (< 1).
- Two eccentricites $e_{\rm in}$ and $e_{\rm out}$ are uniformly sampled between 0.01 and 0.99. We reject systems that do not satisfy the condition for hierarchy: $a_{\rm in}(1 + e_{\rm in}) > a_{\rm out}(1 - e_{\rm out})$.
- Six orbital angles are isotropically sampled: two inclinations $i_{\rm in}$ and $i_{\rm out}$ (range from 0 to π and uniform in their cosines), longitudes of ascending node $\Omega_{\rm in}$ and $\Omega_{\rm out}$ (uniform from 0 to 2π), and two arguments of periapsis $\omega_{\rm in}$ and $\omega_{\rm out}$ (uniform from 0 to 2π).

In addition to these parameters, two true anomalies (orbital phases) $\nu_{\rm in}$ and $\nu_{\rm out}$ need to be specified. We first construct a 4×4 grid of variations of mean anomalies $M_{\rm in}$ and $M_{\rm out}$, keeping all the other parameters fixed,

$$M_{\rm in}, M_{\rm out} \in \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}.$$

We then convert the mean anomalies to true anomalies by numerically solving Kepler's equation. These angles are identical for circular orbits but not for eccentric orbits.

Finally, we consider another variation per system by perturbing the nominal inner semimajor axis ratios,

$$a_{\rm in}^{\rm pert} = a_{\rm in}^{\rm nominal} + 10^{-6},$$

while all other orbital parameters and masses are held fixed.

This results in a total of $2 \times 4 \times 4 = 32$ variations per base triple system. Using zero-based indexing, the variation index is

$$v = 16s + 4p_{\rm in} + p_{\rm out},$$

where $s \in \{0, 1\}$ denotes the nominal or perturbed inner-semimajor-axis setting, and $p_{\rm in}, p_{\rm out} \in \{0, 1, 2, 3\}$ index the nominal orbital phases. Consequently, variations $v = 0, \ldots, 15$ form the nominal 4×4 phase grid, while variations $v = 16, \ldots, 31$ repeat the same phase configurations after perturbing the inner semimajor axis.

This construction serves two complementary purposes. The 4×4 phase grid captures the dependence of the long-term dynamics on orbital phase, while the paired semimajor-axis perturbations provide controlled probes of sensitivity to extremely small changes in the initial conditions. In a chaotic regime, such nearly identical initial configurations can evolve toward substantially different long-term outcomes.

A.2 Datasets and Statistics

Of the total 160,000 individual triple systems generated, we ignore 64 of them since their N-body simulations do not reach completion within our maximum time limit of 5 hr per system.

The dataset we offer has the following characteristics:

- The full features included in the initial condition and evolving trajectory are listed in Table 3 and Table 4
- Each ensemble of system has 32 perturbations near the stability boundary that share similar initial conditions. The exact differences are shown in Appendix A.1
- It has more systems near astrophysical stability boundary, and the detailed distribution is shown in Figure 2

Table 2: Construction of the 32 initial-condition variations for each hierarchical triple. Variations 1–16 span a 4×4 grid of nominal inner- and outer-orbit phases. Variations 17–32 repeat the same phase grid after increasing the inner semimajor axis by 10^{-6}.

Nominal j	Perturbed j	Φ_{in}	Φ_{out}
1	17	0	0
2	18	0	$\pi/2$
3	19	0	π
4	20	0	$3\pi/2$
5	21	$\pi/2$	0
6	22	$\pi/2$	$\pi/2$
7	23	$\pi/2$	π
8	24	$\pi/2$	$3\pi/2$
9	25	π	0
10	26	π	$\pi/2$
11	27	π	π
12	28	π	$3\pi/2$
13	29	$3\pi/2$	0
14	30	$3\pi/2$	$\pi/2$
15	31	$3\pi/2$	π
16	32	$3\pi/2$	$3\pi/2$

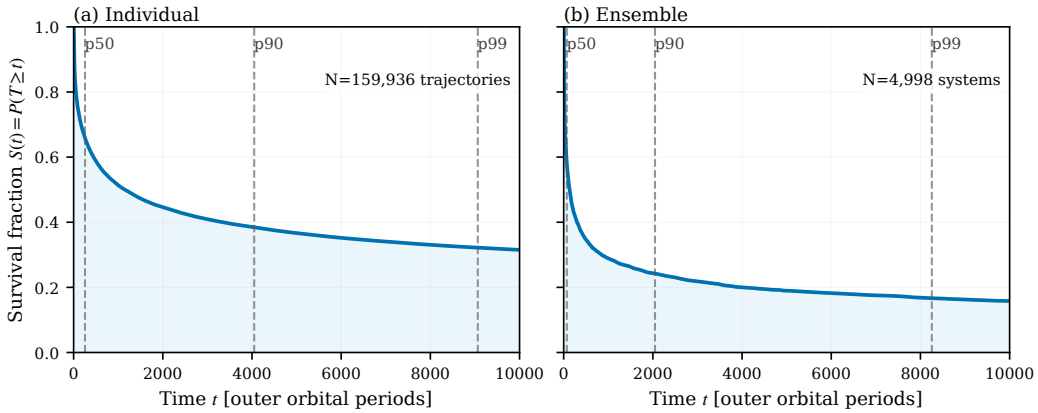


Figure 2: Distribution of the dataset at the individual and ensemble levels.

Dynamic summary of the prefix Dynamical summary statistics extracted from each trajectory channel over the observed prefix (Table 5). For a channel $x(t)$ with prefix length T, the same set of statistics is computed independently for all 15 orbital channels. Dynamic summary statistics which are used as features for some tasks.

A.3 Data Splits

To prevent information leakage between closely related initial-condition variations, all train–validation–test partitions are constructed at the underlying-ensemble level. In particular, individual system generated from the same ensemble of systems are never assigned to different splits.

Stability prediction splits. The initial-condition-based prediction tasks and the short-term-prefix-based stability prediction tasks share the same e-level split. After removing incomplete systems, the dataset contains 4,998 ensemble of systems and 159,936 individual systems, with 32 initial-condition variations per ensemble of systems. We use a $70\%/15\%/15\%$ train–validation–test split with random seed 42, resulting in 3,498 training ensemble of systems, 750 validation ensemble of systems, and 750 test ensemble of systems.

For ensemble-level prediction, each ensemble of systems constitutes one sample. For individual-level stability classification, the individual trajectories are treated as separate samples only after

Table 3: Fields describing the initial conditions of the hierarchical triple systems.

Key	Physical meaning
`aratios`	Ratio of the inner to outer semimajor axes, $a_{\rm in}/a_{\rm out}$, characterizing the hierarchical structure of the system.
`mratios`	Two mass ratios, m_1/m_2 and $m_3/(m_1+m_2)$. The former describes the mass ratio of the inner binary, while the latter measures the mass of the outer body relative to the total mass of the inner binary.
`eccs`	Eccentricities of the inner and outer orbits, $(e_{\rm in}, e_{\rm out})$.
`incs`	Inclinations of the inner and outer orbits with respect to the reference plane, $(i_{\rm in}, i_{\rm out})$. These are not the mutual inclination; the mutual inclination must be computed from the angular-momentum directions of the two orbits.
`APs`	Arguments of periapsis of the inner and outer orbits, $(\omega_{\rm in}, \omega_{\rm out})$.
`LANs`	Longitudes of ascending node of the inner and outer orbits, $(\Omega_{\rm in}, \Omega_{\rm out})$.
`tAnos`	True anomalies of the inner and outer orbits, $(\nu_{\rm in}, \nu_{\rm out})$, describing the initial orbital phases.

Table 4: Fields storing the time evolution of the simulated hierarchical triple systems.

Key	Physical meaning
`times_complete`	Time grid for each trajectory. The dataset spans from $t = 0$ to 10,000 initial outer-orbital periods.
`aratios_complete`	Time evolution of the instantaneous osculating semimajor-axis ratio, $a_{\rm in}(t)/a_{\rm out}(t)$.
`mratios_complete`	Time series of the mass ratios m_1/m_2 and $m_3/(m_1+m_2)$. For point-mass integrations without mass transfer, these quantities remain constant.
`eccs_complete`	Instantaneous osculating eccentricities of the inner and outer orbits, $(e_{\rm in}(t), e_{\rm out}(t))$.
`incs_complete`	Instantaneous inclinations of the inner and outer orbits with respect to the reference plane, $(i_{\rm in}(t), i_{\rm out}(t))$.
`APs_complete`	Instantaneous arguments of periapsis of the inner and outer orbits, $(\omega_{\rm in}(t), \omega_{\rm out}(t))$.
`LANs_complete`	Instantaneous longitudes of ascending node of the inner and outer orbits, $(\Omega_{\rm in}(t), \Omega_{\rm out}(t))$.
`tAnos_complete`	Instantaneous true anomalies of the inner and outer orbits, $(\nu_{\rm in}(t), \nu_{\rm out}(t))$, describing the orbital phases. These quantities may become undefined when the orbital elements become degenerate or the system becomes unstable.

Table 5: Dynamic summary statistics which are used as features for some tasks.

Statistic	Definition
Mean	$\frac{1}{T}\sum_{t=1}^{T} x(t)$
Standard deviation	$\sqrt{\frac{1}{T}\sum_{t=1}^{T}(x(t)-\bar{x})^2}$
Minimum	$\min_{1\le t\le T} x(t)$
Maximum	$\max_{1\le t\le T} x(t)$
Range	$\max_t x(t) - \min_t x(t)$
First value	$x(1)$
Last value	$x(T)$
Net change (Δ)	$x(T) - x(1)$
Linear slope	Slope coefficient obtained by least-squares linear regression of $x(t)$ on t

the ensemble-level split has been determined. The resulting individual system counts are therefore 111,936, 24,000, and 24,000 for the training, validation, and test sets, respectively. Varying the observed prefix length or the number of initial-condition variations used by a model does not alter the underlying system assignment.

Table 6: Ensemble-level split used for the initial-condition-based and short-term-prefix-based stability prediction experiments. All 32 initial-condition variations associated with a ensemble of systems are kept in the same partition.

Split	Ensembles	Individuals
Train	3,498	111,936
Validation	750	24,000
Test	750	24,000

Trajectory forecasting split. Trajectory forecasting requires a separate split because a trajectory must contain a sufficiently long valid segment to construct a complete context–prediction window. Under our default forecasting protocol, each example contains a 512-step context followed by a 128-step prediction horizon. We therefore first retain only trajectories for which at least one complete 640-step forecasting window can be constructed.

After this completion-aware filtering, the forecasting subset contains 64,597 individual systems from 3,847 ensembles of systems. These eligible systems are then independently divided using the same $70\%/15\%/15\%$ proportions and random seed 42, yielding 2,692 training ensembles of systems, 577 validation ensembles of systems, and 578 test ensembles of systems. The split remains strictly ensemble-disjoint, although not every eligible ensemble of system necessarily retains all 32 individual system after forecasting-window filtering.

Table 7: Ensemble-level split used for trajectory forecasting after completion-aware filtering under the $512 \rightarrow 128$ forecasting protocol.

Split	Ensembles	Individuals
Train	2,692	44,988
Validation	577	9,961
Test	578	9,648
Total	3,847	64,597

Because forecasting windows extracted from the same trajectory can strongly overlap, we subsample candidate windows during training and evaluation. During training, at most eight windows are randomly sampled from each trajectory per epoch. Validation and one-shot forecasting evaluation use up to four fixed windows per trajectory, while long-horizon rolling evaluation uses up to two starting points per trajectory. Feature normalization statistics are estimated exclusively from finite, pre-instability observations in the training split.

A.4 Evaluation Metric Definitions

We evaluate all models on held-out systems. Unless otherwise specified, reported metrics are computed on the test split after model and hyperparameter selection using only the training and validation splits.

Binary classification. For a binary target $y_i \in \{0, 1\}$, let $\hat{p}_i$ denote the predicted probability of the positive class and $\hat{y}_i = \mathbb{I}[\hat{p}_i \geq \tau]$ the corresponding binary prediction. Accuracy and F1 are defined as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \tag{4}$$

and

$$\text{F1} = \frac{2\,\text{Precision}\,\text{Recall}}{\text{Precision} + \text{Recall}}, \qquad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{5}$$

When a threshold τ is optimized, it is selected exclusively on the validation split by maximizing F1 and is then kept fixed for test evaluation. We additionally report AUROC and AUPRC, which are threshold-independent and are computed from $\hat{p}_i$. The positive class is explicitly indicated in each result table; stability is treated as the positive class in tables reporting "stable F1." Because F1 and AUPRC depend on the positive-class convention, stable-positive and unstable-positive values are not directly interchangeable.

For individual-level classification, each individual perturbation is treated as one example. For ensemble-level evaluation, probabilities from the V_i perturbations of ensemble of systems i are averaged:

$$\hat{p}_i^{\text{ensemble}} = \frac{1}{V_i} \sum_{j=1}^{V_i} \hat{p}_{ij}. \tag{6}$$

The corresponding ensemble target is the observed fraction of stable perturbations. In the complete ensemble experiments, $V_i = 32$.

Regression. For targets y_i and predictions $\hat{y}_i$, we report

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i|, \tag{7}$$

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2, \tag{8}$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2}, \tag{9}$$

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{10}$$

Pearson correlation measures linear agreement between predictions and targets, while Spearman correlation, where reported, measures agreement between their ranks. Predictions of bounded quantities such as stable fractions are clipped to their valid range $[0, 1]$ before evaluation.

Survival prediction. Let T_{ij} be the event time of perturbation j in ensemble of system i, and let

$$Y_{ijk} = \mathbb{I}[T_{ij} > t_k] \tag{11}$$

be its empirical survival indicator at evaluation horizon t_k. Given a predicted survival probability $\hat{S}_{ij}(t_k)$, the predicted ensemble-level survival fraction is

$$\hat{S}_i^{\text{ensemble}}(t_k) = \frac{1}{V_i} \sum_{j=1}^{V_i} \hat{S}_{ij}(t_k). \tag{12}$$

The horizon-wise ensemble MAE is

$$\text{MAE}(t_k) = \frac{1}{N} \sum_{i=1}^{N} \left| \hat{S}_i^{\text{ensemble}}(t_k) - S_i^{\text{ensemble}}(t_k) \right|, \tag{13}$$

and the overall survival MAE is its mean across evaluation horizons.

The Brier score at horizon t_k is evaluated at the perturbation level:

$$\text{BS}(t_k) = \frac{1}{\sum_i V_i} \sum_{i=1}^{N} \sum_{j=1}^{V_i} \left(\hat{S}_{ij}(t_k) - Y_{ijk} \right)^2. \tag{14}$$

Our discrete integrated Brier score is

$$\text{IBS} = \frac{1}{K} \sum_{k=1}^{K} \text{BS}(t_k). \tag{15}$$

The reported experiments use $K = 20$ logarithmically spaced time bins between $t = 5$ and $t = 10{,}000$. This is an equally weighted discrete approximation: each ensemble of systems, each perturbations, and each time bins receive equal weight. Event indicators are observed over the complete simulation interval; indvidual systems that remain bound at the terminal horizon are treated as surviving through the evaluation window. Consequently, the reported IBS does not apply inverse-probability-of- censoring weighting (IPCW).

Trajectory forecasting. Continuous trajectory channels are standardized using statistics estimated from the training split only:

$$z_{itc} = \frac{x_{itc} - \mu_c^{\text{train}}}{\sigma_c^{\text{train}}}. \tag{16}$$

We compute MSE, MAE, and RMSE in this standardized feature space. Scalar forecasting metrics are ensemble-balanced: errors are first averaged over forecast horizons and channels within each prediction window, then over windows from the same individual system, over and finally over ensemble of systems. This prevents systems contributing different numbers of windows from receiving different weights.

Pearson correlation is computed within prediction windows and aggregated using the same ensemble-balanced hierarchy. Horizon-wise errors are obtained by holding the forecast step fixed and averaging over channels, individual systems, and ensembles of systems. For autoregressive rollouts, these metrics are reported separately for successive forecast blocks to characterize accumulated error.

Angular variables are represented by sine–cosine pairs. In addition to their standardized forecasting errors, we measure violation of the unit-circle constraint using

$$E_{\text{circle}} = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \left| \hat{s}_a^2 + \hat{c}_a^2 - 1 \right|, \tag{17}$$

where $(\hat{s}_a, \hat{c}_a)$ is a predicted sine–cosine pair and $\mathcal{A}$ contains all predicted angular pairs, horizons, and evaluated examples. Lower values indicate that the predicted representation remains closer to the unit circle. This metric measures geometric consistency of the angular representation rather than angular displacement from the target.

A.5 Full Results

A.5.1 Task A: Stability Prediction from Initial Conditions

Table 8: Binary stability classification from initial physical conditions. Stability is treated as the positive class.

Method	Accuracy	Precision	Recall	Stable F1	AUROC	AUPRC
Mardling–Aarseth criterion [21]	0.6187	0.2875	**0.9496**	0.4414	0.7529	0.2810
Vynatheya et al. criterion [36]	0.6600	0.3046	0.8908	0.4540	0.7536	0.2887
Vynatheya et al. MLP classifier [36, 37]	0.7267	0.3547	0.8824	0.5060	**0.9086**	**0.7184**
MLP	0.7507	0.3629	0.7563	0.4905	0.8416	0.5139
XGBoost	**0.8053**	**0.4308**	0.7059	**0.5350**	0.8525	0.5089

Table 9: Stable-fraction regression from initial physical conditions.

Method	MSE	RMSE	MAE	R^2
Mardling-Aarseth criterion [21]	0.2488	0.4988	0.3073	-0.5653
Vynatheya et al. criterion [36]	0.2268	0.4763	0.2853	-0.4269
Vynatheya et al. MLP classifier [36, 37]	0.1056	0.3249	**0.2176**	0.3359
MLP	0.0901	0.3001	0.2247	0.4335
XGBoost	**0.0884**	**0.2972**	0.2330	**0.4442**

A.5.2 Task B: Stability Prediction from Early Dynamics

Results are shown in Table 10.

Task b ablation: At-risk evaluation. We mainly carry out two ablations on both individual and ensemble level to remove the systems that have already got unstable within the prefix. Results are shown in Table 11 and Table 12.

A.5.3 Task C: Multi-horizon Survival Prediction

Results are shown in Table 13.

Table 10: Performance of different trajectory-prefix representations. Individual-level predictions are additionally aggregated over the 32 variations of each ensemble of system to evaluate ensemble-level stable-fraction prediction. Stability is treated as the positive class.

			Individual-level classification				Ensemble-level prediction		
Prefix length	Representation	Model	Acc.	Stable F1	AUROC	AUPRC	MAE	MSE	R^2
96	PANDA encoder	Linear	0.8640	0.8070	0.9467	0.8911	0.1031	0.0256	0.8390
	Chronos-2 encoder	Linear	0.8623	0.8067	0.9531	0.9083	0.1112	0.0281	0.8230
	Dynamical summaries	XGBoost	**0.9000**	**0.8404**	**0.9649**	**0.9319**	**0.0703**	**0.0153**	**0.9040**
	`orbital15`	SVM	0.7493	0.5800	0.8135	0.6166	0.2373	0.0927	0.4165
	`orbital15`	CNN	0.7937	0.6092	0.8888	0.7526	0.1759	0.0786	0.5056
	`orbital15`	TCN	0.8167	0.7402	0.8971	0.7523	0.1570	0.0581	0.6347
512	PANDA encoder	Linear	0.9217	0.8845	0.9786	0.9550	0.0534	0.0071	0.9554
	Chronos-2 encoder	Linear	0.9209	0.8848	0.9791	0.9552	0.0614	0.0087	0.9452
	Dynamical summaries	XGBoost	**0.9329**	**0.8960**	**0.9851**	**0.9673**	**0.0398**	**0.0051**	**0.9676**
	Raw `orbital15`	SVM	0.7590	0.5918	0.8282	0.6232	0.2184	0.0818	0.4855
	Raw `orbital15`	CNN	0.8839	0.8205	0.9470	0.8543	0.0985	0.0278	0.8249
	Raw `orbital15`	TCN	0.8973	0.8411	0.9593	0.8900	0.0765	0.0177	0.8886

Table 11: At-risk ablation for ensemble-level stability prediction. Evaluation is restricted to systems that remain bound through the specified time t_{risk}. All models use XGBoost. Lower is better for both metrics.

At-risk cohort	Representation	Ensembles	MAE	MSE
$prefix > 96$	Initial conditions	257	0.3432	0.1551
	Dynamical summary, prefix 96	257	**0.1103**	**0.0269**
$prefix > 512$	Initial conditions	174	0.3985	0.1940
	Dynamical summaries, prefix 96	174	0.0897	0.0187
	Dynamical summaries, prefix 512	174	**0.0520**	**0.0079**

At-risk cohorts include only individual systems that remain bound up to the specified time.

Table 12: At-risk individual-level stability classification. Only individual systems that remain bound through the specified at-risk time t_{risk} are retained. Stability is treated as the positive class.

At-risk cohort	Prefix	Representation	Predictor	Individuals	Stable %	Acc.	Stable F1	AUROC	AUPRC
$Prefix > 96$	96	PANDA	Linear probe	14,192	54.04	0.7709	0.8075	0.8716	0.8923
		Chronos-2	Linear probe			0.7672	0.8067	0.8848	0.9088
		Dynamical summaries	XGBoost			**0.8309**	**0.8404**	**0.9123**	**0.9320**
$Prefix > 512$	96	PANDA	Linear probe	10,189	75.27	0.7877	0.8631	0.8172	0.9301
		Chronos-2	Linear probe			0.7885	**0.8649**	0.8341	0.9398
		Dynamical summaries	XGBoost			**0.7999**	0.8611	**0.8655**	**0.9537**
	512	PANDA	Linear probe			0.8180	0.8859	0.8787	0.9578
		Chronos-2	Linear probe			0.8147	0.8854	0.8781	0.9573
		Dynamical summaries	XGBoost			**0.8421**	**0.8961**	**0.9040**	**0.9674**

For an at-risk time t_P, evaluation retains only individual systems satisfying $T > t_P$ and predicts whether they remain stable through $H = 10{,}000$. The strict inequality excludes individual systems that become unbound exactly at the observation endpoint. All classification metrics use a fixed probability threshold of 0.5, with stability treated as the positive class. Stable % denotes the positive-class prevalence.

Table 13: Multi-horizon survival prediction performance. All learned representations use the same MLP hazard head.

Method	Aggregation	Overall MAE	Final MAE	IBS
Baselines				
Train-mean	–	0.3099	0.3510	0.1315
Raw `orbital15` CNN	–	0.0910	0.1728	0.0752
Raw `orbital15` Dynamical summaries	–	0.0670	0.1475	0.0632
Raw `orbital15` TCN	–	0.0662	0.1564	0.0606
Pretrained time-series representations				
PANDA	Mean/std pooling	0.0602	0.1035	0.0550
	Channel-attention pooling	0.0459	0.0892	0.0486
	Direct	0.0427	**0.0819**	0.0459
Chronos-2	Mean/std pooling	0.0476	0.0939	0.0480
	Channel-attention pooling	0.0414	0.0887	0.0426
	Direct	**0.0376**	0.0826	**0.0416**

Overall MAE is averaged over all 20 prediction horizons, whereas Final MAE evaluates only the final horizon. IBS denotes the integrated Brier score.

A.5.4 Long-horizon Trajectory Forecasting

Table 14: Trajectory forecasting performance.

Model	Setting	MSE (z)	MAE (z)	Pearson	Unit-circle Error
Chronos-2 [1]	Zero-shot	0.2331	**0.2282**	0.8580	**0.2265**
PANDA [16]	Zero-shot	0.2217	0.2371	0.8625	0.2862
iTransformer [20]	Supervised	**0.2123**	0.2429	**0.8673**	0.3152
TOTO2 [15]	Zero-shot	0.2262	0.2303	0.8601	0.2310
TiRex2 [25]	Zero-shot	0.2422	0.2408	0.8508	0.2277

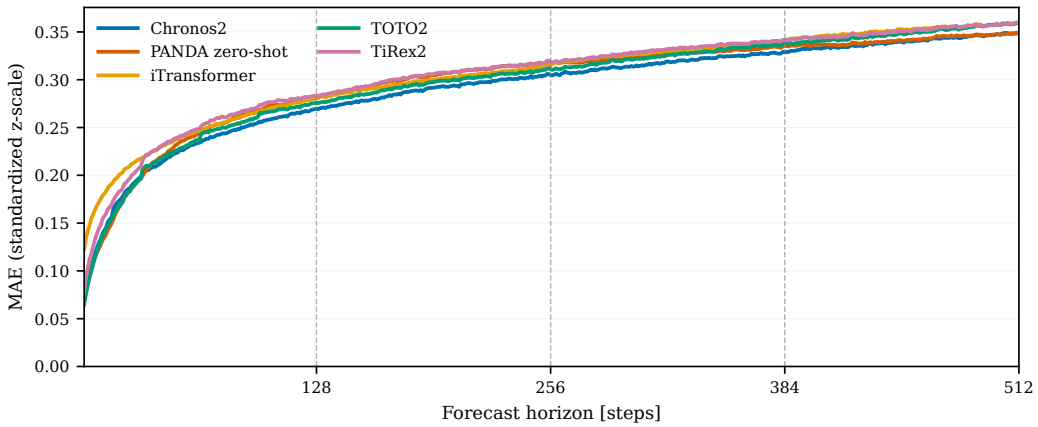


Figure 3: Error Accumulation during Autoregressive Long-Horizon Rollouts.